作业（共3页，第一页）

- 集合论（选做、要求掌握）：
  1. 教材第一章 1,
  2. 教材第一章 2.
- 概率模型：
  3. Six cups and saucers come in pairs: there are two cups and saucers which are red, two white, and two with stars on. If the cups are placed randomly onto the saucers (one each), find the probability that no cup is upon a saucer of the same pattern.
  4. 教材第一章 6,
  5. 小清、小华、小紫三人轮流抛一枚硬币，第一次抛出正面朝上的人获胜。试验的样本空间可以定义为：$S = \{1,01,001,0001,\cdots\}$. 请根据$S$，写出下列事件的具体表达：(i) 小清胜，记作事件A；(ii)小华胜，记作事件B；(iii) $(A \cup B)^c$.

作业（共3页，第二页）

- 概率性质：

6. 某商店售卖一种盲盒，每盒内有一个印有24节气之一的书签。试求购买n盒这种盲盒而集齐24节气书签的概率。(Hint:古典概型、定义合适的事件来表达目标事件、可用课上已学的概率性质；写出最终含n的计算式即可，无需化简)

- （选做、要求掌握）：

7. 教材第一章 9,
8. 教材第一章 10

9. 证明概率性质第7条的第一个不等式
10. You are given that at least one of the events $A_r, 1 \leq r \leq n$, is certain to occur, but certainly no more than two occur. If $\mathbb{P}(A_r) = p$, $\forall\, 1 \leq r \leq n$ and $\mathbb{P}(A_r \cap A_s) = q, \forall\, 1 \leq r \neq s \leq n$, show that $p \geq 1/n$ and $q \leq 2/n$.

作业（共3页，第三页）

- 随机抽样：

11. （选做）请证明有放回无序抽样的结论$\binom{M+n-1}{n}$.

提示：1) 可以使用结论$\binom{m}{n}=\sum_{k=n-1}^{m-1}\binom{k}{n-1}$。

2) 可以固定M和n中的一个去思考，需要选择固定哪一个更便捷。

- 注：所有选做题都不计分，以后的作业也是。

思考题

- “可列”可加性，可否放宽至“任意”可加性?
- 任意一个样本空间的子集都可以分配概率么?

具体题目

SECTION 1 . 1 . Sets

Problem 1. Consider rolling a six-sided die. Let A be the set of outcomes where the roll is an even number. Let B be the set of outcomes where the roll is greater than 3. Calculate and compare the sets on both sides of De Morgan's laws

$$(A \cup B)^c = A^c \cap B^c. \qquad (A \cap B)^c = A^c \cup B^c.$$

Problem 2. Let A and B be two sets.

(a) Show that

$$A^c = (A^c \cap B) \cup (A^c \cap B^c), \qquad B^c = (A \cap B^c) \cup (A^c \cap B^c).$$

(b) Show that

$$(A \cap B)^c = (A^c \cap B) \cup (A^c \cap B^c) \cup (A \cap B^c).$$

(c) Consider rolling a fair six-sided die. Let A be the set of outcomes where the roll is an odd number. Let B be the set of outcomes where the roll is less than 4. Calculate the sets on both sides of the equality in part (b), and verify that the equality holds.

Problem 6. A six-sided die is loaded in a way that each even face is twice as likely as each odd face. All even faces are equally likely, as are all odd faces. Construct a probabilistic model for a single roll of this die and find the probability that the outcome is less than 4.

Problem 9. A partition of the sample space Ω is a collection of disjoint events $S_1, \ldots, S_n$ such that $\Omega = \cup_{i=1}^{n} S_i$.

(a) Show that for any event A, we have

$$\mathbf{P}(A) = \sum_{i=1}^{n} \mathbf{P}(A \cap S_i).$$

(b) Use part (a) to show that for any events A, B, and C, we have

$$\mathbf{P}(A) = \mathbf{P}(A \cap B) + \mathbf{P}(A \cap C) + \mathbf{P}\big(A \cap B^c \cap C^c\big) - \mathbf{P}(A \cap B \cap C).$$

Problem 10. Show the formula

$$\mathbf{P}\big((A \cap B^c) \cup (A^c \cap B)\big) = \mathbf{P}(A) + \mathbf{P}(B) - 2\mathbf{P}(A \cap B),$$

which gives the probability that exactly one of the events A and B will occur. [Compare with the formula $\mathbf{P}(A \cup B) = \mathbf{P}(A) + \mathbf{P}(B) - \mathbf{P}(A \cap B)$, which gives the probability that at least one of the events A and B will occur.]

证明概率性质7的第一个不等式

7. (Bonferroni's inequality) 如果 $A_i \in \mathscr{F}$, $i = 1, \ldots, n$, 则

$$\mathbb{P}\Big(\bigcup_{i=1}^{n} A_i\Big) \geq \sum_{1 \leq i \leq n} \mathbb{P}(A_i) - \sum_{1 \leq i < j \leq n} \mathbb{P}(A_i \cap A_j).$$

(Kounias's inequality)

$$\mathbb{P}\Big(\bigcup_{i=1}^{n} A_i\Big) \leq \min_k \left\{ \sum_{1 \leq i \leq n} \mathbb{P}(A_i) - \sum_{i: i \neq k} \mathbb{P}(A_i \cap A_k) \right\}.$$